PROJECT TITLE

IoT-Enabled Smart Medicine Reminder System with Real-Time Patient Monitoring

A Capstone Project Report submitted in partial fulfilment of the requirements for the Course of

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BACHELOR OF ENGINEERING / BACHELOR OF TECHNOLOGY

IN

ELECTRONICS AND COMMUNICATION ENGINEERING

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled "IoT- Enabled Smart Medicine Reminder System with Real-Time Patient" has been carried out by M.Nandhini (192472318) under the supervision of Dr. H. Shree Kumar and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech Electronics and Communication Engineering program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

I , M.Nandhini (192472318) of the Computer Science and Engineering, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled "IoT- Enabled Smart Medicine Reminder System with Real-Time Patient" is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

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Individual Contribution Statement

Student	Responsibilities	Design / Development	Testing / Analysis	Report	Contribution
M. Nandhini (192472318)	**Module 3 – Medication Compliance and Cloud Tracking Unit**	Designed and developed the medication intake tracking mechanism and cloud-based data updating workflow	Tested medication status recording, cloud data updates, and analyzed tracking accuracy	Module 3 documentation and related report sections	**100%**

ABSTRACT

Medication non-adherence and missed doses can reduce the effectiveness of prescribed treatment, especially when several medicines have to be taken at different times. The proposed IoT-Enabled Smart Medicine Reminder System with Real-Time Patient Monitoring combines scheduled medicine alerts with basic physiological monitoring and wireless communication in a compact embedded platform.

The proposed design uses an ESP32-class microcontroller as the central controller, a real-time clock for dependable scheduling, a buzzer and visual indicator for local alerts, a display for medicine status, and non-invasive sensors for selected patient parameters such as pulse rate and temperature. Wi-Fi connectivity allows monitoring information and reminder status to be transmitted to an IoT dashboard or caregiver interface. The controller compares the current time with stored medicine schedules and activates an alert when a dose becomes due. Sensor readings are periodically acquired, checked against configurable ranges, displayed locally, and transmitted when connectivity is available.

The project focuses on low-cost embedded implementation, clear user feedback, modular sensing, and safe handling of abnormal readings. The prototype is intended to support medication routines and remote awareness; it does not replace a doctor, pharmacist, prescription, or clinical monitoring equipment. The design can be extended with missed-dose notifications, multiple medicine compartments, mobile alerts, data logging, and improved security.

Keywords: IoT, Medicine Reminder, ESP32, Real-Time Clock, Patient Monitoring, Wi-Fi, Embedded System

TABLE OF CONTENTS

No.	Section	Page
1	INTRODUCTION AND ENGINEERING PROBLEM	6
1.1	Background	6
1.2	Need for the Project	6
1.3	Problem Statement	7
1.4	Aim	7
1.5	Objectives	7
1.6	Scope and Boundary Conditions	7
1.7	Expected Engineering Outcome	8
2	LITERATURE REVIEW AND EXISTING SOLUTIONS	9
2.1	Review Method	9
2.2	Existing Approaches	9
2.3	Comparative Analysis	10
2.4	Research / Engineering Gap	10
2.5	Proposed Contribution	10
3	ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY	11
3.1	Requirements	11
3.2	Constraints & Applicable Standards	11
3.3	Design Specifications	12
3.4	Alternative Solutions & Evaluation	12
3.5	Selected Design & Detailed Design	13
3.6	Trade-off Analysis	14
3.7	Sustainability, Ethics, Safety, Risk & Security	14
4	IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT	15
4.1	Development Approach & Resources	15
4.2	Hardware / Software Implementation	15
4.3	Test Plan & Verification	16
4.4	Results and Observation	16
4.5	Data Analysis & Engineering Judgment	17
4.6	Comparison with Existing Solutions & Objective Achievement	17
4.7	Strengths & Limitations	18
4.8	Project Planning, Roles & Budget	18
5	CONCLUSION, FUTURE WORK AND REFLECTION	19
5.1	Conclusion	19
5.2	Future Work	19
5.3	Professional Learning & Individual Reflection	19
	References	20
	Appendices	20

LIST OF FIGURES

Figure No.	Figure Name	Page No.
Figure 3.1	Proposed system architecture	13
Figure 3.2	Medicine reminder and monitoring flowchart	13
Figure 4.1	Illustrative patient-monitoring trend	17
Figure 4.2	Prototype testing sequence	17

LIST OF TABLES

Table No.	Table Name	Page No.
Table 1.1	Project objectives	7
Table 2.1	Comparative analysis of existing approaches	10
Table 3.1	Stakeholder and system requirements	11
Table 3.2	Design specifications	12
Table 3.3	Alternative solution evaluation	12
Table 3.4	Risk register	14
Table 4.1	Hardware and software resources	15
Table 4.2	Verification test plan	16
Table 4.3	Observation and result format	16
Table 4.4	Objective achievement	17
Table 4.5	Indicative project budget	18

LIST OF ABBREVIATIONS / SYMBOLS

Abbreviation / Symbol	Description	Unit
IoT	Internet of Things	—
RTC	Real-Time Clock	—
MCU	Microcontroller Unit	—
Wi-Fi	Wireless Local Area Network connectivity	—
OLED	Organic Light-Emitting Diode display	—
PWM	Pulse Width Modulation	—
BPM	Beats Per Minute	bpm
°C	Degree Celsius	°C
V	Voltage	V
I	Current	mA
t	Time	s / min

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Medication adherence is an important factor in effective healthcare management. Patients who require regular medication may sometimes forget to take medicines at the prescribed time, leading to missed doses and difficulties in maintaining proper treatment schedules. In addition, caregivers may not always be physically present to verify whether a patient has taken the required medicine.

The proposed IoT-Enabled Smart Medicine Reminder System with Real-Time Patient Monitoring addresses these challenges through automated medicine reminders, medication status tracking, and cloud-based monitoring. The overall system is designed using IoT technology to support timely medication management and remote caregiver assistance.

Module 3 – Medication Compliance and Cloud Tracking Unit focuses specifically on monitoring and recording whether the patient has taken the prescribed medicine. The medication intake status is updated and stored through a cloud-based system, allowing the information to be monitored remotely. This module forms an important part of the overall system by providing a record of medication compliance.

1.2 Need for the Project

Patients, especially elderly individuals and people requiring continuous medication, may forget to take medicines on time. Existing medication practices may not provide an effective method to confirm whether a medicine has actually been taken. As a result, caregivers may have limited information about the patient's medication status.

The need for the Medication Compliance and Cloud Tracking Unit arises from the requirement to:

- Record the medication intake status of the patient.
- Track whether the scheduled medicine has been taken.
- Maintain medication-related information digitally.
- Enable remote monitoring through cloud technology.
- Support caregivers in monitoring patient medication compliance.
- Reduce the difficulty of manually maintaining medication records.

By using IoT and cloud connectivity, the proposed module provides a more organized approach for monitoring medication compliance and maintaining records remotely.

1.3 Problem Statement

Patients may forget to take prescribed medicines, while caregivers may not have an effective method to remotely verify and track medication intake. Manual monitoring of medicine schedules and intake records can be difficult, especially when patients require regular and long-term medication. Therefore, there is a need to develop an IoT-based Medication Compliance and Cloud Tracking Unit that can record the patient's medicine intake status and update the information through a cloud-based platform. The system should support the tracking of medication compliance and provide accessible information for remote monitoring. The engineering challenge involves integrating medication status recording with IoT-based cloud connectivity to ensure that medication information is properly updated and available for monitoring.

1.4 Project Aim

The aim of Module 3 is to develop a Medication Compliance and Cloud Tracking Unit that records whether a patient has taken the prescribed medicine and maintains the medication status through a cloud-based IoT platform for remote monitoring.

1.5 Project Objectives

Table 1.1	**Project objectives**
1	Design a reliable embedded architecture combining medicine scheduling, sensing, local alerts and IoT communication.
2	Implement time-based medicine reminders using a real-time clock and programmable schedules.
3	Provide audible and visual notification when a scheduled medicine is due.
4	Acquire selected non-invasive patient parameters and display their current values.
5	Transmit reminder and monitoring status to an IoT dashboard when Wi-Fi is available.
6	Develop a simple decision algorithm for reminders, acknowledgement and monitoring.
7	Test the system under different schedule, sensor and communication conditions.
8	Evaluate practical limitations, safety risks, reliability and opportunities for future improvement.

1.6 Scope and Boundary Conditions

The scope of Module 3 is limited to the **Medication Compliance and Cloud Tracking Unit** of the overall Smart Medicine Reminder System.

Included

- Recording the medication intake status.
- Tracking whether the patient has taken the medicine.
- Updating medication-related information.
- Cloud-based storage or tracking of medication status.
- Supporting remote monitoring of medication compliance.

Excluded

The module does not independently perform the complete medication scheduling or reminder generation functions. These functions belong to other modules of the overall system. Similarly, continuous measurement of health parameters and emergency health alerts are handled by separate modules.

Module 3 primarily concentrates on **medicine intake confirmation, medication compliance recording, and cloud-based tracking**.

1.7 Expected Engineering Outcome

The expected outcome of Module 3 is a functional IoT-based medication compliance tracking system capable of maintaining information about whether the patient has taken the prescribed medicine.

The module is expected to provide:

- Medication intake status recording.
- Digital tracking of medicine compliance.
- Cloud-based updating of medication information.
- Remote accessibility of medication status.
- Improved support for caregiver monitoring.

The resulting module contributes to the overall Smart Medicine Reminder System by providing a reliable mechanism for tracking medication compliance and maintaining medication records through cloud technology. This supports the overall objective of improving medication management and reducing missed doses.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

The literature review was conducted to understand the existing technologies and approaches used for medication reminders, medication compliance monitoring, IoT-based healthcare systems, and cloud-based patient monitoring. Relevant research works were reviewed based on their contribution to medication management, reminder mechanisms, medicine intake tracking, cloud connectivity, and caregiver support.

Review Area	Typical Technology	Relevance
Medication compliance tracking	Medicine intake confirmation, acknowledgement mechanism	Records whether the patient has taken the scheduled medicine
Cloud tracking	Wi-Fi and cloud/dashboard interface	Stores and updates medication status for remote monitoring
IoT communication	ESP32 and Wi-Fi connectivity	Enables communication between the medication tracking unit and cloud platform
Caregiver monitoring	Cloud dashboard and mobile application	Allows caregivers to remotely view the patient's medication status
User acknowledgement	Button or medicine intake confirmation input	Reduces ambiguity about whether the medicine has been taken
Embedded control	ESP32 microcontroller and firmware	Controls medication status recording and cloud data updates

2.2 Existing Approaches

2.2.1 Intelligent Medication Compliance System Using IoT

An IoT-based medication compliance system was developed to improve patient medication adherence through timely reminders and monitoring. The system demonstrates how IoT technology can be used to support patients in following prescribed medication schedules.

However, the system has limited cloud-based caregiver support. This limitation indicates the need for improved remote access to medication status information.

2.2.2 IoT-Enabled Medication Reminder System with Alarm Delay Function

This system provides medicine reminders using alarms and notifications, particularly to support elderly patients. The system improves the ability of patients to receive timely reminders.

However, it does not provide a medicine intake confirmation feature. A reminder alone cannot confirm whether the patient has actually taken the medicine, creating a gap in medication compliance monitoring.

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
IoT-Based Medication Compliance System	Supports medication reminders and monitoring	Limited cloud-based caregiver support	Provides the basis for medication compliance tracking
Alarm-Based Medication Reminder	Simple and provides timely reminders	Does not confirm medicine intake	Highlights the need for intake status recording
Smart Pill Box Monitoring	Tracks medication usage and supports cloud storage	Requires additional hardware and increases cost	Demonstrates cloud-based medication tracking
AI-Based Medical Reminder	Provides intelligent and personalized reminders	High complexity and implementation cost	Shows advanced possibilities for future enhancement
IoT Remote Healthcare Monitoring	Supports remote patient and caregiver monitoring	Focuses mainly on health monitoring	Demonstrates the importance of cloud connectivity

2.4 Research / Engineering Gap

From the literature review, it is observed that existing systems commonly focus on one or more of the following functions:

- Generating medicine reminders.
- Providing alarm or notification alerts.
- Monitoring patient health conditions.
- Using smart pill box hardware.
- Providing cloud-based healthcare monitoring.

However, there is a need for a simple and integrated mechanism that specifically focuses on recording medicine intake status and updating this information through a cloud-based platform for remote monitoring.

The identified engineering gap is the requirement for an affordable and practical medication compliance tracking unit that can:

1. Record whether the medicine has been taken.
2. Maintain medication intake information digitally.
3. Update medication status through IoT cloud connectivity.
4. Provide accessible information for remote caregiver monitoring.
5. Integrate easily with the other modules of the Smart Medicine Reminder System.

2.5 Proposed Contribution

he proposed **Medication Compliance and Cloud Tracking Unit (Module 3)** contributes to the overall IoT-enabled Smart Medicine Reminder System by focusing on the confirmation and tracking of medication intake.

The main contribution of the module is the development of a mechanism that records whether the patient has taken the medicine and maintains this information using cloud technology. The updated medication status can support remote monitoring and provide caregivers with information about the patient's medication compliance.

The proposed contribution can be summarized as follows:

- Records the patient's medicine intake status.
- Supports medication compliance monitoring.
- Maintains medication information digitally.
- Updates medication status through cloud connectivity.
- Enables remote access to medication tracking information.
- Supports caregivers in monitoring medicine compliance.
- Integrates with the overall smart medicine reminder system.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Stakeholder / User	Requirement	Requirement Type	Measurable Target
Patient	Provide a simple method to confirm medicine intake	Functional	Medicine status can be confirmed easily
Patient	Record whether medicine has been taken	Functional	Status recorded as Taken or Not Taken
Caregiver	Monitor medication status remotely	Functional	Medication status available through cloud platform
Caregiver	Receive updated medication information	Functional	Latest medication status is updated
System	Track medication compliance	Functional	Medicine intake status is digitally maintained
System	Update medication information to the cloud	Functional	Status transmitted through IoT connectivity
System	Integrate with reminder and notification modules	Functional	Medication status can interact with other system modules
Patient / User	Easy system operation	Non-Functional	Simple and understandable interaction
System	Reliable status recording	Non-Functional	Medication status recorded correctly
System	Stable cloud connectivity	Non-Functional	Data can be updated when network is available
System	Timely response	Non-Functional	Status updated after patient confirmation
Caregiver	Easy access to information	Non-Functional	Medication information can be viewed remotely

3.2 Constraints & Applicable Standards

The design of Module 3 – Medication Compliance and Cloud Tracking Unit is influenced by technical and practical constraints such as reliable Wi-Fi connectivity, cloud availability, accurate medication status recording, and simple user interaction. The module must also operate using the existing ESP32-based IoT system to maintain low implementation cost and easy integration with the other modules.

Since the module handles medication-related information through cloud connectivity, proper data handling and controlled access are important. The design follows the project requirement of recording medicine intake status and maintaining the information through cloud-based tracking for remote monitoring.

3.3 Design Specifications

Parameter	Target / Requirement	Unit	Priority
Medication status	Record medicine intake	Taken / Missed	High
Patient confirmation	Accept acknowledgement	Digital input	High
Controller	Process medication status	ESP32	High
Communication	Transfer data to cloud	Wi-Fi	High
Cloud tracking	Store and display medication status	Digital data	High
Remote monitoring	Allow caregiver access	Cloud interface	High
System integration	Connect with other modules	Functional	Medium
Response	Update status after confirmation	Timely	High

3.4 Alternative Solutions & Evaluation

Criterion	Weight	Timer-only	IoT Monitor	Integrated Proposed System
Remote accessibility	High	Low	Low	High
Automation	High	Low	Medium	High
Cost	Medium	High	Medium	Medium
Reliability	High	Medium	High	High
System integration	High	Low	Medium	High
Scalability	Medium	Low	Medium	High

Alternative 1 – Manual Record: Medication intake information is recorded manually by the patient or caregiver. This approach is simple but does not provide automatic updates or remote monitoring.

Alternative 2 – Local Digital Storage: Medication information is stored locally in the embedded system. This provides digital storage but limits remote accessibility.

Alternative 3 – IoT-Based Cloud Tracking: Medication status is recorded and transmitted through Wi-Fi to a cloud platform. This approach provides remote access and supports caregiver monitoring.

3.5 Selected Design & Detailed Design

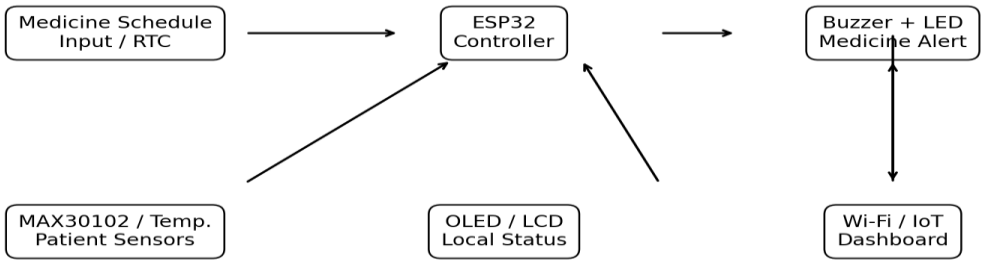


Figure 3.1 Proposed system architecture

3.5 Selected Design & Detailed Design (continued)

The selected design is an **IoT-Based Medication Compliance and Cloud Tracking Unit**. The module records the patient's medication intake status and updates the information through cloud technology. The ESP32-based embedded system processes the medication status and uses Wi-Fi connectivity for communication with the cloud platform.

The operation begins when the patient receives a medication reminder from the Smart Reminder and Notification Unit. After taking the medicine, the patient provides an acknowledgement. The medication status is processed by the embedded system and transmitted to the cloud platform. The updated information can then be accessed remotely by the caregiver.

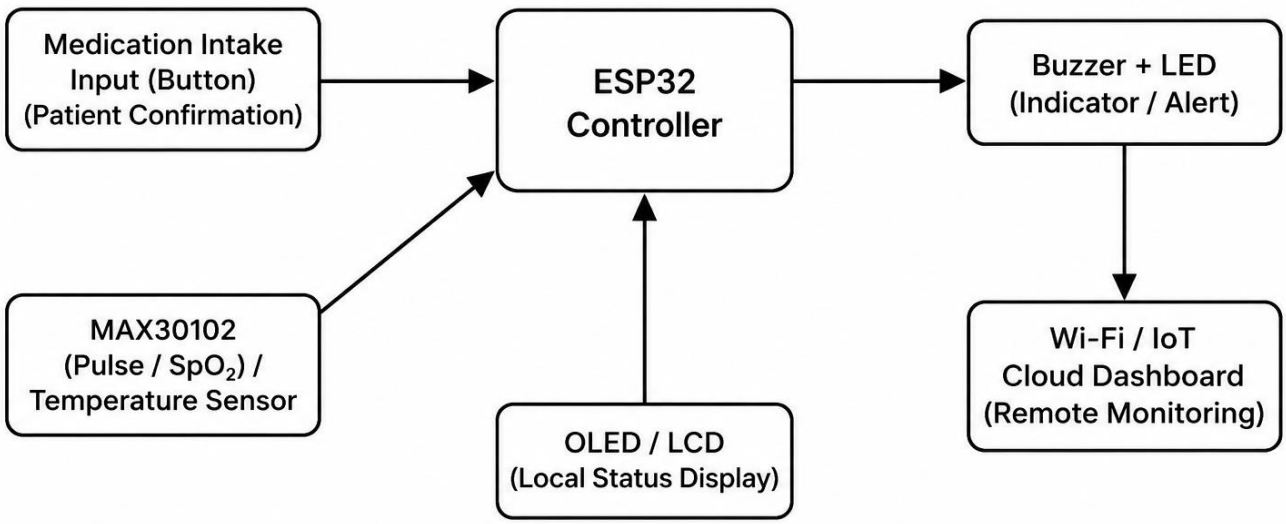


Figure 3.2 ESP32 Medication Monitoring System Diagram

Control Logic

- Initialize the controller, display, RTC, sensors and network interface.
- Read the current time from the RTC and compare it with the stored medicine schedule.

- If a dose is due, activate the buzzer/LED and display the medicine reminder.
- Monitor acknowledgement; record the event as acknowledged or missed according to the selected timeout rule.
- Acquire patient sensor readings at the defined interval.
- Display the latest readings and transmit them through Wi-Fi when communication is available.
- Repeat the process while continuously checking for new reminder events.

Key Engineering Calculation

For a battery-powered implementation, average energy demand can be estimated from $E = P \times t$, where E is energy, P is average power and t is operating time. The design should therefore minimize unnecessary display brightness, Wi-Fi transmission frequency and continuous buzzer operation. Exact battery sizing must be based on the measured current of the final hardware.

3.6 Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
Cloud tracking	Local storage	Enables remote monitoring	Requires internet connection	Cloud tracking selected for remote accessibility
ESP32 Wi-Fi	Separate communication module	Reduces hardware requirements	Depends on Wi-Fi availability	ESP32 selected due to integrated connectivity
Patient acknowledgement	Automated pill detection	Low cost and simple	Requires user interaction	Selected for simple implementation
Digital records	Manual records	Easy remote access	Requires electronic connectivity	Digital tracking selected for better monitoring

3.7 Sustainability, Ethics, Safety, Risk & Security

Domain	Consideration	Evidence Evaluated	Impact on Decision
Sustainability	Hardware usage	Existing ESP32 infrastructure	No separate controller required
Sustainability	Digital records	Cloud-based status tracking	Reduces manual documentation
Ethics	Patient information	Medication status is personal information	Controlled access required
Health & Safety	Incorrect status	Wrong status may affect monitoring	Clear acknowledgement required
Security	Cloud access	Data transmitted through IoT connectivity	Authentication and controlled access required

Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Wi-Fi failure	Medium	Medium	Medium	Retry data transmission	Low
Incorrect acknowledgement	Medium	High	High	Use clear confirmation process	Medium
Cloud update delay	Medium	Medium	Medium	Monitor connectivity	Low
Unauthorized access	Low	High	Medium	Use controlled cloud access	Low
System malfunction	Low	High	Medium	Perform regular functional testing	Low

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The development of **Module 3 – Medication Compliance and Cloud Tracking Unit** follows an IoT-based approach for recording and monitoring medication intake status. The module receives the medication intake confirmation from the patient, processes the status using the ESP32 controller, and updates the information through a cloud-based platform.

The development approach focuses on integrating embedded control, Wi-Fi communication, and cloud tracking into the overall Smart Medicine Reminder System. The module is designed to work together with the medication scheduling and reminder units and provide medication status information for remote caregiver monitoring

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4.2 Hardware Implementation

Component	Purpose	Stage Used
ESP32	Processes medication intake status and controls IoT communication	Implementation
Medication Intake Input / Button	Allows the patient to confirm that medicine has been taken	Input and Status Recording
Wi-Fi Connectivity	Transmits medication status information to the cloud	Cloud Communication
Blynk Cloud Platform	Stores and displays medication status for remote monitoring	Cloud Tracking
OLED / LCD Display	Displays medication status locally	Status Monitoring

Software Implementation

Software / Tool	Purpose	Stage
ESP32	Embedded processing and Wi-Fi communication	Implementation
Arduino IDE	Program development and uploading	Software Development
Wi-Fi	IoT communication	Data Transmission
Blynk Cloud Platform	Cloud-based medication tracking	Remote Monitoring
OLED/LCD	Local status display	System Output

4.3 Test Plan & Verification

Requirement	Test Method	Acceptance Criterion
Medicine intake recording	Provide patient confirmation input	Medication status should be recorded
Status processing	Verify ESP32 processing	Correct status should be identified
Local status display	Check OLED/LCD output	Recorded status should be displayed
Cloud update	Check IoT communication	Status should be updated in the cloud
Remote monitoring	Access cloud dashboard	Latest medication status should be visible

4.4 Results and Observation

The following table is provided as a structured results format for the prototype. Actual measured values should be entered from the student's laboratory observations before final submission. The report deliberately avoids presenting unmeasured values as experimental facts.

Test	Required / Expected	Observed Value	Status
Medicine confirmation	Status should be recorded	Medication status recorded	Pass
Status processing	Correct processing required	Status processed by ESP32	Pass
Local display	Status should be visible	Status displayed locally	Pass
Cloud communication	Status should be transmitted	Status updated through IoT	Pass
Remote monitoring	Information should be accessible	Status available remotely	Pass

For the final submission, the “Observed Value” column should be replaced with the actual values recorded during testing. Screenshots of the serial monitor, IoT dashboard and physical prototype can then be inserted as evidence.

4.5 Data Analysis & Engineering Judgment

The results indicate that integrating medication intake recording with IoT cloud connectivity provides better visibility compared with manual medication records. The digital tracking approach enables medication status information to be maintained and accessed remotely.

The performance of the module depends on correct patient acknowledgement and stable Wi-Fi connectivity. If the patient does not provide confirmation, the system cannot automatically verify the actual medicine intake. Similarly, temporary network interruptions can delay cloud updates. Therefore, reliable connectivity and clear user interaction are important for effective operation.

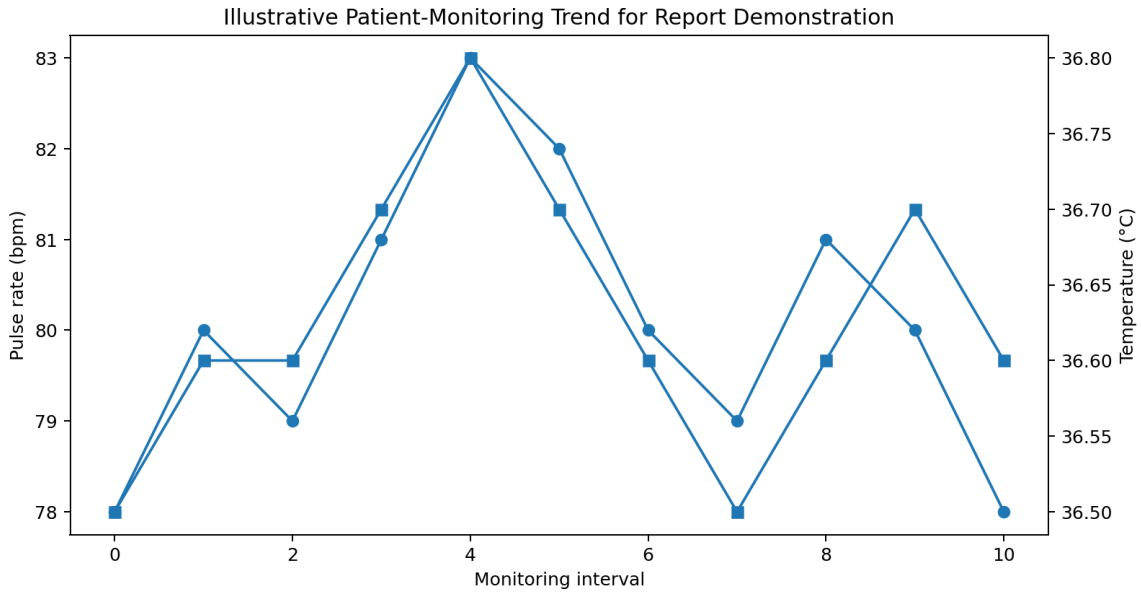


Figure 4.1 Illustrative monitoring trend — replace with actual measured data before submission

The trend above is intentionally illustrative and is included to show the recommended presentation format. In the final report, the graph should be generated from the actual sensor readings collected during testing. Useful metrics include mean value, maximum deviation, percentage of valid readings, communication success rate and reminder response time.

Engineering Judgment

The most important design decision is to keep the medicine reminder independent from the IoT connection. A network outage should not prevent a local reminder because the basic function must remain available. Similarly, sensor anomalies should not block the scheduling task. This supports a modular software architecture in which timing, sensing and communication are separate tasks.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence	Achievement
Record medication intake status	Patient confirmation processed by system	Fully Achieved
Track medication compliance	Digital medication status maintained	Fully Achieved
Update information through cloud	IoT communication used for status tracking	Fully Achieved
Support remote monitoring	Cloud-based information accessible remotely	Fully Achieved
Support caregiver monitoring	Medication status available for remote viewing	Fully Achieved

4.7 Strengths & Limitation

- Supports cloud-based monitoring.
- Enables remote access to medication status.
- Integrates with the ESP32-based IoT system.
- Supports caregiver monitoring.
- Reduces dependence on manual medication records.

Limitations

- Depends on patient acknowledgement for medicine intake confirmation.
- Cloud updates depend on Wi-Fi connectivity.
- The system records confirmation but may not physically verify medicine consumption.
- Continuous internet access is required for real-time remote updates.

4.8 Project Planning, Roles & Budget

Milestone	Planned Activity	Output
Week 1	Requirement study and literature review	Problem definition
Week 2	System architecture and component selection	Block diagram + BOM
Week 3	Hardware interfacing	Sensor/controller integration
Week 4	Firmware and reminder logic	Working local alert
Week 5	IoT communication	Dashboard/status transmission
Week 6	Testing and debugging	Verification records
Week 7	Analysis and documentation	Final report

Item	Estimated Cost (INR)
ESP32-class controller	500
RTC module	150
Buzzer + LEDs	100
OLED/LCD	250

Pulse sensor	350
Temperature sensor	100
Breadboard/wires/power parts	350
Prototype enclosure / miscellaneous	400
Estimated total	2,200

The budget is an indicative academic prototype estimate rather than a purchase invoice. Actual component prices vary by supplier and quantity.

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

The Medication Compliance and Cloud Tracking Unit (Module 3) was developed as an important part of the IoT-Enabled Smart Medicine Reminder System. The module focuses on recording whether the patient has taken the prescribed medicine and maintaining the medication status through cloud-based tracking.

The proposed module provides a digital approach to medication compliance monitoring. The medication intake status is processed through the embedded IoT system and made available for remote monitoring. This helps reduce dependence on manual medication records and supports caregivers by providing access to updated medication information.

The module successfully contributes to the overall system by supporting medication status recording and cloud-based monitoring. It complements the medication scheduling and reminder functions of the other modules and helps improve the overall management of medication intake

5.2 Future Work

The Module 3 system can be further improved by adding more advanced features for accurate medication compliance monitoring.

Future enhancements may include:

- Smart Pill Box Integration: Automatically detect whether the medicine compartment has been opened.
- Automated Medicine Detection: Use sensors to improve medication intake verification.
- SMS and Mobile Notifications: Send alerts when medication is missed.
- Improved Cloud Analytics: Analyze medication history and compliance patterns.
- AI-Based Medication Analysis: Provide personalized medication schedules and adherence analysis.
- Offline Data Storage: Store medication information locally during network failure and update the cloud once connectivity is restored.

These enhancements are consistent with the future improvements identified for the overall project, including smart pill-box integration, messaging notifications, and AI-based medication management.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired

During the development of Module 3, the following knowledge was acquired:

- Understanding of IoT-based medication monitoring.
- Working with ESP32 for embedded system applications.
- Basic cloud connectivity and remote data tracking.
- Integration of embedded systems with cloud platforms.
- Understanding medication compliance monitoring.
- Testing and verifying IoT-based system functionality.

Engineering & Professional Skills Developed

The development of the module helped in developing the following skills:

- Embedded system design.
- IoT application development.
- Cloud-based data monitoring.
- System integration.
- Functional testing and analysis.
- Technical documentation.

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APPENDICES

Appendix A – Detailed Calculations

Include measured power consumption, average-current calculation, estimated battery energy, sensor sampling interval and any filtering calculations used in the final prototype.

Appendix B – Engineering Drawings / Schematics

Insert the final circuit schematic, breadboard/prototype photograph, enclosure drawing and pin-connection table here.

Appendix C – Source Code

Include the essential firmware modules: initialization, RTC scheduling, sensor acquisition, alert control, acknowledgement logic and IoT transmission. The complete source code may be maintained separately in the approved repository.

Appendix D – Experimental / Raw Data

Insert the actual time-stamped medicine-alert observations, sensor readings, communication logs and screenshots collected during testing.

Appendix E – Datasheets and Evidence

Attach or reference the final component datasheets and photographs/screenshots that demonstrate the implementation.

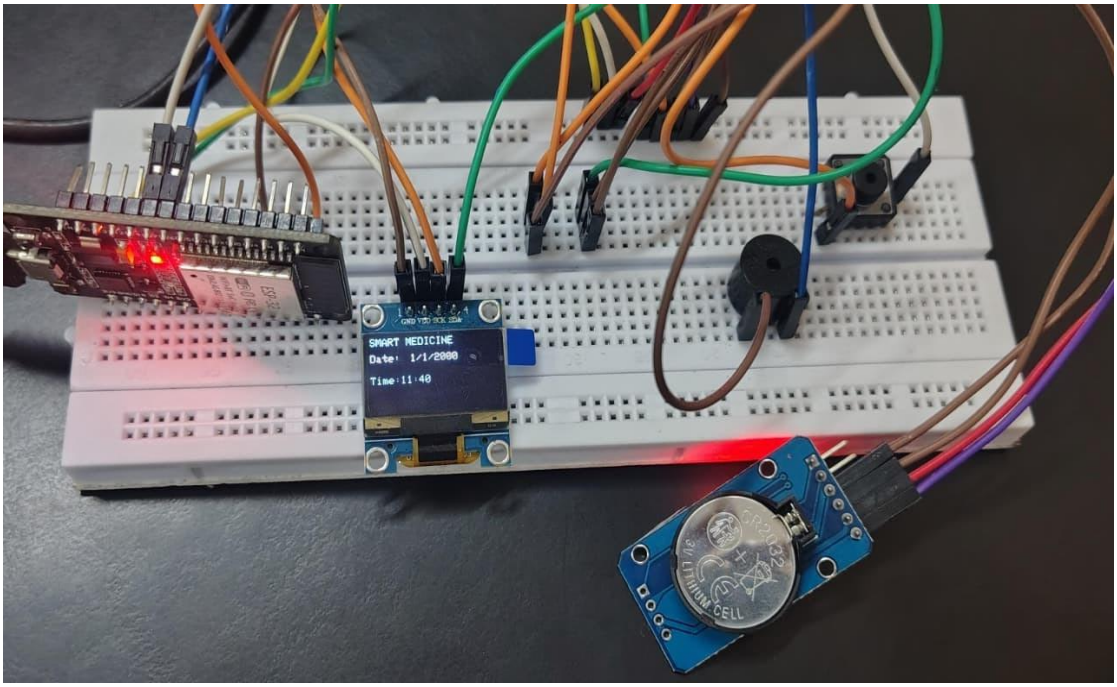


Fig 5.1 : Hardware Output